pfi

# Python for Fixed Income

COURSE 8 · IMPLEMENTING AGENTIC AI

# Document Tools

WEEK 2 · DAY 2

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# Today: documents as tools

### Six tools, every text with its source line

`statement_dates`

`statement_text`

`list_sections`

`read_section`

`eightk_section`

`search_indenture`

### Today

1. Documents as tools
2. The body, not the contents
3. Search as a tool
4. Quotes and citations, checked
5. NOT IN TEXT, through a tool
6. An 8-K section, described
7. What a long result costs

**Every model response in the notebook is a saved response, replayed. A live run can differ.**

# The rules for documents

## An FOMC statement: what it states, never a policy view

| Allowed names | Never called |
| --- | --- |
| the statement states | hawkish or dovish |
| a sentence in one and not the other | a policy view |
| the date of the statement | what the Committee will do |

## An indenture: what it states, never a judgement of the issuer

| Allowed names | Never called |
| --- | --- |
| the indenture states | loose, tight, weak, strong |
| extracted value, with its quote | covenant-lite or protective |
| passed the source check | a credit view or recommendation |

# Documents as tools

**statement_text("2026-09-16", 300): the text, the cut marker, the source line**

```
The Committee decided to raise the target range for the federal funds rate by 1/4 percentage po...

... [cut: 843 characters in all]
```

Source: FOMC post-meeting statement of 2026-09-16, Board of Governors of the Federal Reserve System, https:...

```
A date with no statement: {'error': 'no statement dated 2026-09-17 in the file'}
```

## a dict

the same shape from every tool

## max_chars

the model chooses how much text

## Source:

the last line of every text

**A quote the model copies always arrives with the name of its document.**

# The body, not the table of contents

### Every heading textkit's pattern reads as Section 4.12 in Boyd's text

```
character   1,996:  Section 4.12. Limitation on Indebtedness 56

character 198,454:  Section 4.12. Limitation on Indebtedness.
```

**find_sections keeps the heading at 198,454, the last one, of 107 sections**

**56**

a page number: the contents line

**11,868**

characters in the body of 4.12

**2.0 to 1.0**

the ratio, in the body's first sentence

Source: Boyd Gaming Corporation, indenture for the 4.750% Senior Notes due 2031, Exhibit 4.1 to the 8-K filed 2021-06-08, accession 0001193125-21-185495, Section 4.12.

**read_section reads from the body's heading: the section's words, not its page number.**

# Search as a tool: ranks, never scores

**search_indenture("limitation on liens", 3) over 454 chunks of 384 numbers**

```
rank 1  Section 4.14   Section 4.14. Limitation on Liens. The Company shall not, and shall no
rank 2  Section 4.14   Liens (or any portion thereof) described in the definition of "Permitt
rank 3  Section 1.01   arrangements for the sale of goods entered into by that Person or any
```

Source: Boyd Gaming Corporation, indenture for the 4.750% Senior Notes due 2031, Exhibit 4.1 to the 8-K filed 2021-06-08, accession 0001193125-21-185495, Section 4.14.

## no score

last digits move with threads

## in the key

a result is part of every later request

## > 1e-6

every gap the builder checked

**A rank is how close the words sit. It is not a judgement of the text.**

# Quotes and citations, checked

### citations_ok (Course 7's judge) and quotes_found, on the saved answers

| run | accepted | citations | numbers | missing | quotes found |
|---|---|---|---|---|---|
| fomc | False | False | False | no section read [True] | |
| section | True | True | True | [] | [False] |
| search | True | True | True | [] | [True] |

moved to [Section 4.10]: {'numbers_ok': True, 'citations_ok': False, 'accepted': False, 'missing': []}

"the Company's Consolidated Fixed Charge Coverage Ratio would have been at least 2.0 to 1.0, after gi...

**section: the quote wrote 'Pro Forma Basis' in single quotes; the section has double. A changed quote mark is a changed quote.**

# NOT IN TEXT, through a tool

### What the agent did, step by step

```
1  read_section  {"indenture":"boyd_2021","number":"4.12","max_chars":6000} tool_calls 768 31
```

### The saved answer

```
NOT IN TEXT

citations_ok False: no section cited; numbers_ok True
```

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 2 requests, 3,002 prompt and 35 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

**The reply the prompt asks for. The judge never accepts NOT IN TEXT: count those replies apart.**

# An 8-K section, described

### Item 2.03, as its filer wrote it: the saved answer's quote

```
The section states: "On June 15, 2021, Public Service Company of New Hampshire, doing business as...

$350,000,000 aggregate principal amount of its 2.20% First Mortgage Bonds, Series V, due 2031
```

The answer wrote the inner quotes as \"Company\"; the quote check reads them as plain quotes.

```
quotes found [True, True]; numbers no tool returned []
```

Source: PUBLIC SERVICE CO OF NEW HAMPSHIRE, Form 8-K filed 2021-06-21, accession 0001104659-21-082904, Item 2.03.

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 2 requests, 2,013 prompt and 237 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

**What the filer wrote, under the item it chose. Not a view on the bonds, not a view on the issuer.**

# What a long tool result costs

**Prompt tokens by request, in the saved runs of the lesson**

| run | tokens by request | added | tool result characters |
|---|---|---|---|
| fomc | 768, 1061 | 293 | 1109 |
| section | 789, 2255 | 1466 | 6368 |
| search | 789, 1173, 1411 | 384, 238 | 1372, 639 |
| yield | 768, 2234 | 1466 | 6368 |
| eightk | 789, 1224 | 435 | 1377 |

**1466**

section run: 4.12 at max_chars 6000

**1466**

yield run: 4.12 at max_chars 6000

**238**

search run: 4.14 at max_chars 300

**Every later request carries the whole transcript: a long result is paid for again at every step.**

# Excel and Git

**Excel side by side (tested in Excel)**

| | | | | |
|---|---|---|---|---|
| `=ISNUMBER(MATCH(B2,C2:G2,0))` | `B 4.12 (text)` | `C 4.12` | `(section)` | TRUE |
| `=ISNUMBER(MATCH(B3,C3:G3,0))` | `B 4.14 (text)` | `C 4.14` | `(search)` | TRUE |
| `=ISNUMBER(MATCH(B5,C5:G5,0))` | `B 4.12 typed as a number` | | | FALSE |
| `=C8-C7` | `prompt tokens, section run` | | | 1466 |
| `=C11-C10` | `prompt tokens, yield run` | | | 1466 |

**Excel against Python: 4 of 4 rows the same**

```
...at least 100, for example [-6000.-]{+2000.+}
first request's key: 6f5f45311685 saved, 32ceba4ebfca edited
```

**git diff --word-diff: one word of a tool description, and every request key changes. 40 passed.**

# The AI check

The docstring is the prompt. The agent code that comes back runs and looks right.

1

## Docstring

Read the section from the body: its first sentence, never the contents line.

→

2

## Code back

Written for the course in the style of an AI answer. Read it.

→

3

## Test first

The text for 4.12 must hold Incur any Indebtedness. Run it: it fails.

→

4

## Fix

Let textkit.find_sections find it: the last heading, the body's.

**Today: text.find returns the first heading, the table of contents line, so the reader returns no covenant.**

*Code that runs is not the same as code that is right.*

# What you can do now

- Documents as tools, every text with its source line
- Read a section from the body, not the contents
- Search as a tool: ranks and texts, never scores
- citations_ok and quotes_found; read the missing list
- NOT IN TEXT through a tool; an 8-K described

**Week 2, Day 3** MCP: the same tools, served by a server